Aug-Lag Landscape: Original (75 bases) vs State Aggregation (fewer bases)
Fewer LP variables ⇒ fewer basis regions ⇒ smoother landscape (ρ=10000)

1e7

$L(\lambda) = -r_{true}^T y^*(\lambda) + \rho\|y_{true} - H_{true} y^*\|^2$

State Agg (M=10, 198 bases)
State Agg (M=20, 257 bases)
State Agg (M=30, 292 bases)
$\lambda_{true} = 10.0$

Gradient: state aggregation produces smoother, more informative gradients

1e6

$\partial L/\partial\lambda$ (finite diff)

DFL gradient (S=56)
State Agg gradient (M=10)
State Agg gradient (M=20)
State Agg gradient (M=30)

λ_{pred}